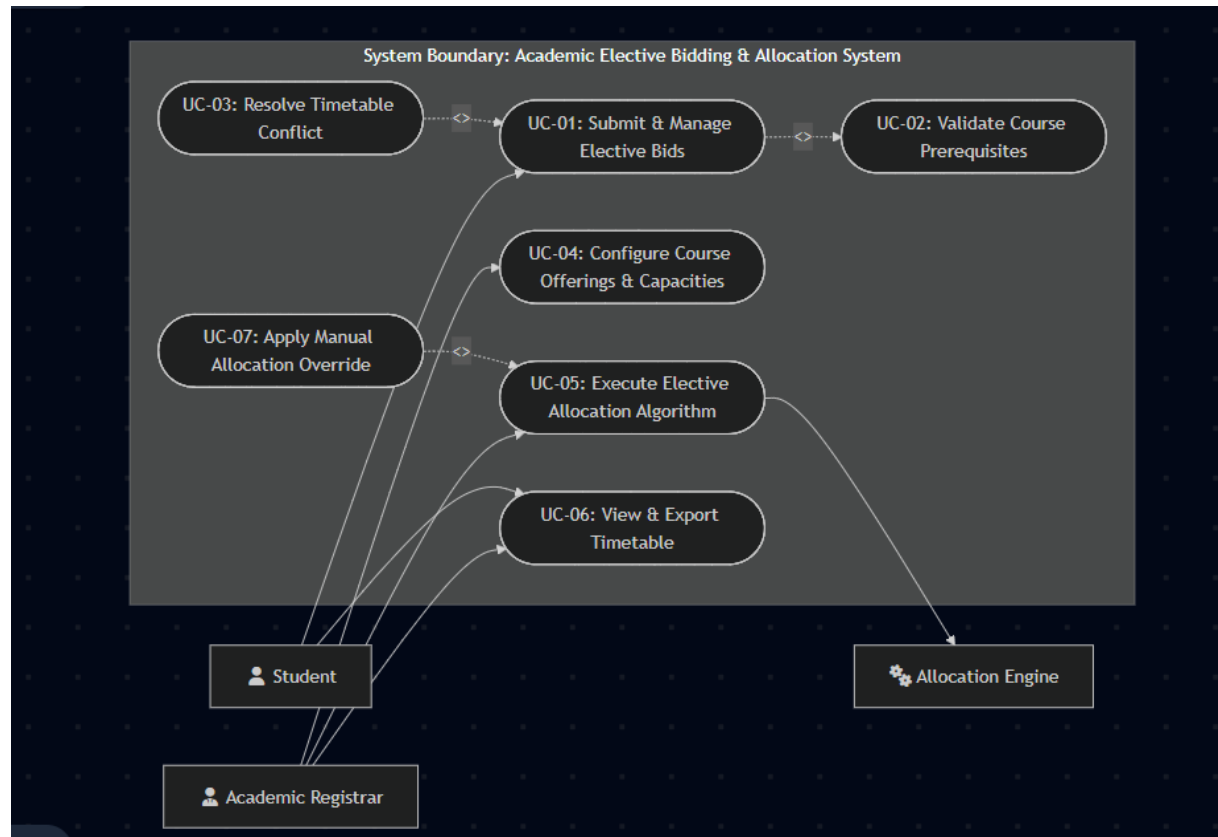


Academic Elective Bidding & Allocation System (Problem Statement #05)

Deliverable 2: UML Use-Case Diagram & Model Architecture



Actors Identified (3)	Primary Use Cases (7)	UML Stereotype Relationships
• Student: Initiates bid point distribution, views allocations. • Academic Registrar: Configures courses, runs solver, oversees rosters. • Allocation Engine: Automated background solver applying optimization constraints.	• UC-01: Submit & Manage Elective Bids • UC-02: Validate Course Prerequisites • UC-03: Resolve Timetable Conflict • UC-04: Configure Course Offerings & Capacities • UC-05: Execute Elective Allocation Algorithm • UC-06: View & Export Timetable • UC-07: Apply Manual Allocation Override	• «include»: UC-01 <i>includes</i> UC-02 (Prerequisites checked for all bid submissions). • «extend»: UC-03 <i>extends</i> UC-01 (Triggered conditionally if chosen courses conflict). UC-07 <i>extends</i> UC-05 (Triggered optionally for registrar manual adjustments).